

# Exact TBR–Cost Pareto Frontier for a Small Toroidal Breeding-Blanket Instance: Software Stack, Model, and Brute-Force Verification

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## Abstract

This report documents the construction and full brute-force verification of a small ( $I = 4$  sectors,  $K = 4$  candidate material recipes,  $4^4 = 256$  total configurations) toroidal fusion breeding-blanket neutronics model, built as the planned small-instance verification case for Research Question 3 (RQ3) of an NSF Engineering Research Initiation proposal on surrogate-screened combinatorial search for PDE-constrained design problems. We describe the open-source neutronics software stack (OpenMC with DAGMC support, Paramak, CadQuery, `cad_to_dagmc`), the specific geometry-construction approach that proved numerically robust after several CAD boolean-operation failures, the tritium-breeding-ratio (TBR) and closed-form material-cost objective functions, and the results of exhaustively evaluating all 256 configurations on a single workstation. The resulting true Pareto frontier (25 non-dominated points out of 256) is reported in full, together with the per-configuration raw simulation output, to serve as a ground-truth benchmark against which a surrogate-screened search algorithm can later be directly checked for optimality rather than merely for exceeding a literature-reported target. All code, configuration files, and raw per-call results are released alongside this report for independent verification.

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# 1 Introduction and Motivation

Design problems for nuclear power system components increasingly require choosing among a large number of discrete combinatorial options against an objective and constraints with no closed form: evaluating a single candidate requires a computationally expensive numerical solve. The PI’s completed preliminary work applied a surrogate-screened combinatorial search framework to a BWR  $10 \times 10$  nuclear fuel assembly design problem, reducing the number of expensive DRAGON5 lattice-physics evaluations from a literature-reported budget of 36,000 per case to 44 total across two problem variants [1]. A subsequent NSF ERI proposal (RQ3) proposes validating the same framework on a structurally different second engineering domain: discrete fusion breeding-blanket material layout design, evaluated via Monte Carlo neutron transport simulation in OpenMC.

The proposal’s Section 3.3.2 specifies a small, brute-force-checkable verification instance before attempting the full-scale  $I = 80$  WCLL-style problem: a reduced blanket with  $I = 4$  toroidal sectors and  $K = 4$  candidate material recipes per sector, giving  $4^4 = 256$  total configurations — small enough to enumerate exhaustively on a single workstation, and therefore capable of providing a *true* optimum (or, in the multi-objective case addressed here, a true Pareto frontier) against which any later approximate search can be checked for optimality directly, not merely for exceeding an external target.

This report documents the complete construction of that small instance: software installation, geometry model, materials, source and tally definitions, the closed-form cost model, and the full 256-configuration enumeration results. It is intended as a companion artifact to the NSF ERI proposal and as a request for expert feedback on the neutronics modeling choices before scaling to the full  $I = 80$  case.

# 2 Software Stack

All work was performed on a single Windows 11 workstation running Ubuntu 24.04 LTS under WSL2, with the WSL distribution relocated to a secondary drive to accommodate the nuclear data library. Table 1 summarizes the installed software stack, all obtained from the `conda-forge` channel except where noted.

Two installation details are worth recording for reproducibility. First, the DAGMC-enabled OpenMC build must be requested explicitly via the conda build-string match `openmc=0.15.3=dagmc*nompi*`; the default `openmc` package resolves to a DAGMC-free, CSG-only build. Second, the PyPI `cad_to_dagmc` package pulls in a pip-distributed `gmsh` wheel with a pre-compiled binary that requires `libXft.so.2`,

Table 1: Software stack (all conda-forge except where noted)

| Component                    | Version                                                 |
|------------------------------|---------------------------------------------------------|
| OpenMC (DAGMC-enabled build) | 0.15.3                                                  |
| DAGMC                        | 3.2.4                                                   |
| MOAB                         | 5.6.0                                                   |
| Paramak                      | 0.9.11                                                  |
| CadQuery                     | 2.8.0                                                   |
| cad_to_dagmc (PyPI)          | 0.14.1                                                  |
| gmsh / python-gmsh           | 4.15.2                                                  |
| Nuclear data library         | ENDF/B-VIII.0 (official pre-generated HDF5, openmc.org) |

which is absent from a minimal WSL Ubuntu install and raises `OSError: libXft.so.2: cannot open shared object file`; this was resolved by removing the pip `gmsh` and installing both `gmsh` and `python-gmsh` from conda-forge instead, which bundle their own dependencies.

The full stack was verified end to end in three stages before attempting the sector model: (1) a pure constructive-solid-geometry OpenMC model (a water sphere with a 14.1 MeV point source) to confirm the nuclear-data path and transport engine; (2) a CadQuery-sphere  $\rightarrow$  `cad_to_dagmc`  $\rightarrow$  DAGMC  $\rightarrow$  OpenMC round trip (a tungsten sphere under the same source) to confirm the CAD-to-neutronics bridge, which also produced a physically sensible leakage fraction above unity (1.556) due to  $(n, 2n)$  neutron multiplication in tungsten at 14.1 MeV; and (3) the sector model described below.

### 3 Problem Formulation

Following the proposal’s general combinatorial design formulation, the small instance assigns one of  $K = 4$  candidate material recipes to each of  $I = 4$  discrete blanket sectors:

$$x_{i,k} \in \{0, 1\}, \quad \sum_{k=1}^K x_{i,k} = 1 \quad \forall i \in \{1, \dots, I\},$$

giving  $K^I = 4^4 = 256$  total configurations. The two objectives are:

- **TBR**( $x$ ) — the tritium breeding ratio, defined as tritium atoms produced per source neutron. This is the expensive oracle output, requiring a full OpenMC Monte Carlo neutron-transport solve on the multi-sector geometry.
- **Cost**( $x$ ) =  $\sum_{i,k} x_{i,k} c_{i,k}$  — the enriched  ${}^6\text{Li}$  material cost, a closed-form linear function of the design requiring no oracle call (Section 7).

The recipes correspond to four  ${}^6\text{Li}$  enrichment levels,  $\varepsilon \in \{0.15, 0.30, 0.60, 0.90\}$ , matching the recipe set already established in the parent proposal’s cost analysis for the full-scale  $I = 80$  case.

## 4 Geometry Model

### 4.1 Design intent

The proposal’s Figure 2 describes the small instance as a torus divided into  $I = 4$  sectors going around the main (toroidal) ring, each assigned one recipe independently. This section documents

the geometry actually built, including the sequence of CAD construction failures encountered and the approach that ultimately proved robust, since this is directly relevant to anyone attempting to reproduce or extend the model.

## 4.2 Construction history and failure modes

Four successive geometry-construction strategies were attempted using CadQuery 2.8.0 (the CAD kernel underlying Paramak), in the following order:

1. **Two concentric tori, boolean cut.** An outer torus (tube radius  $r_{\text{out}}$ ) and an inner torus (tube radius  $r_{\text{in}}$ ) were each built via `revolve(360, ...)` of a circular profile, then subtracted (`outer.cut(inner)`) to form a hollow shell. This raised `ValueError: Null TopoDS_Shape object` from the underlying OpenCASCADE kernel, a known failure mode for boolean operations on two perfectly concentric, seam-aligned solids of revolution.
2. **Full shell, quadrant box intersection.** A single annular profile (circle-within-circle) was revolved  $360^\circ$  in one step to avoid the boolean cut above, then intersected with four large axis-aligned boxes (one per quadrant) to obtain the four sectors. This avoided the immediate crash but caused the boolean `intersect()` call to hang indefinitely — confirmed still consuming 100% CPU after 16 minutes with no completion — attributed to OpenCASCADE numerical instability when a box face is exactly tangent to the torus at the quadrant boundary.
3. **Partial ( $90^\circ$ ) revolve of an annular profile.** To avoid both the full- $360^\circ$  boolean seam and the box-intersection tangency, each sector was built directly as its own  $90^\circ$  partial revolve of an annular (circle-in-circle) profile, with no boolean operations at all. This completed quickly but produced a degenerate result: `cad_to_dagmc` reported `Extracted 0 surfaces from 4 solids` and subsequently `IndexError: list index out of range`, indicating CadQuery’s partial-revolve operation did not correctly interpret the circle-in-circle profile as an annulus in this configuration.
4. **Solid (non-hollow) box sectors.** As a minimal pipeline check, four solid cubes were placed around a ring (no revolve, no boolean operations) and carried all the way through OpenMC. This succeeded completely and confirmed the DAGMC/materials/source/tally pipeline was correct, returning a physically sensible (if geometrically crude) result (Section 8).

## 4.3 Final geometry: Paramak `revolved_shape`

Given the repeated OpenCASCADE boolean-operation failures in Section 4.2, the final geometry uses Paramak’s `paramak.revolved_shape()` function, which accepts an explicit list of (`R`, `Z`, `connection`) profile points and revolves them by a specified angle with no boolean operations required. The profile itself uses a *solid* (non-hollow) circular cross-section, approximated by a 16-sided polygon of straight ("`straight`" connection type) segments centered at radius  $R$ :

$$(R_j, Z_j) = (R + r \cos \theta_j, r \sin \theta_j), \quad \theta_j = \frac{2\pi j}{16}, \quad j = 0, \dots, 15,$$

with major radius  $R = 300$  cm and tube radius  $r = 20$  cm. Each of the four sectors is built as an independent  $90^\circ$  revolve of this profile, then rotated into place at azimuthal offsets 0, 90, 180, 270 and assigned to a named CadQuery assembly component (`sector0`–`sector3`). This construction completed in a few seconds and `cad_to_dagmc` correctly extracted 68 surfaces (14,892 mesh vertices) from the four solids.

**Simplification relative to the full-scale design.** This geometry is a *solid* torus segment, not the hollow, multi-layer (first wall / breeder zone / structural fraction) shell described in the parent proposal’s full-scale  $I = 80$  model, and its dimensions ( $R = 300$  cm,  $r = 20$  cm) were chosen for fast turnaround during pipeline verification rather than calibrated to any specific reference device (e.g. WCLL/DEMO). Reintroducing the hollow shell (first wall + breeder zone) using a boolean-free construction (e.g. two independent `revolved_shape` calls exported separately and combined only at the DAGMC-assembly level, never boolean-cut in CadQuery) is the recommended next step before scaling to  $I = 80$ , and is exactly the kind of geometry-fidelity question this report solicits feedback on.

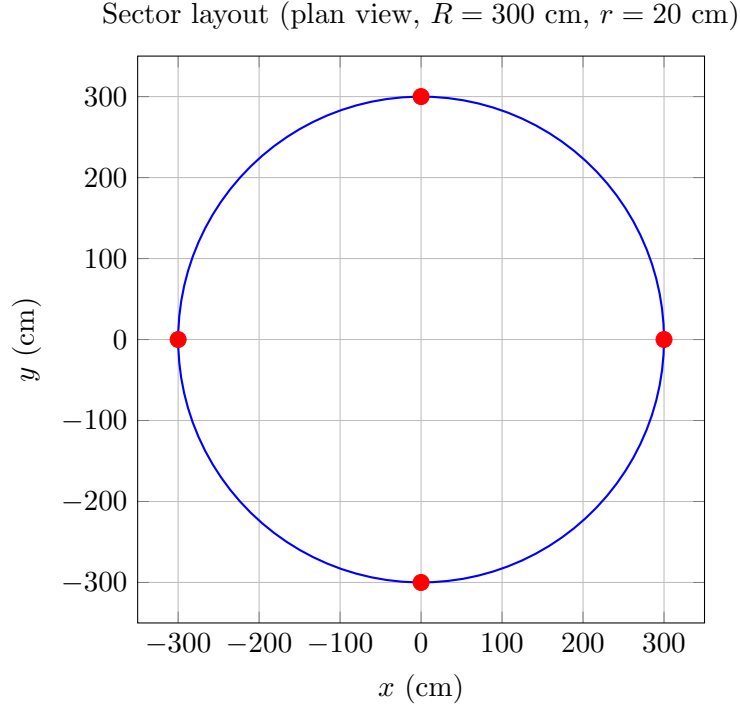


Figure 1: Plan-view schematic of the four sector centers around the  $R = 300$  cm ring (illustrative; actual sectors are  $90^\circ$  toroidal wedges of tube radius 20 cm, not point markers).

## 5 Materials

All four recipes use a PbLi eutectic composition (84.3 at% Pb, 15.7 at% Li), density 10.0 g/cm<sup>3</sup>, differing only in <sup>6</sup>Li enrichment:

```
m = openmc.Material(name=sector_name)
m.add_element('Pb', 84.3, percent_type='ao')
m.add_element('Li', 15.7, percent_type='ao',
              enrichment=eps * 100, enrichment_target='Li6',
              enrichment_type='ao')
m.set_density('g/cm3', 10.0)
```

with  $\varepsilon \in \{0.15, 0.30, 0.60, 0.90\}$  for the four recipes. OpenMC’s automatic natural-abundance expansion resolves Pb into its four naturally-occurring isotopes (Pb204, Pb206, Pb207, Pb208)

and Li into Li6/Li7 at the specified enrichment; all six nuclides were confirmed present in the ENDF/B-VIII.0 library at run time.

## 6 Source and Tally

The D-T fusion plasma is approximated by a monoenergetic (14.1 MeV) ring source at the major radius  $R$ , uniform in toroidal angle and isotropic in direction:

```
space = openmc.stats.CylindricalIndependent(
    r=openmc.stats.Discrete([R], [1.0]),
    phi=openmc.stats.Uniform(0, 2*np.pi),
    z=openmc.stats.Discrete([0.0], [1.0]))
energy = openmc.stats.Discrete([14.1e6], [1.0])
source = openmc.IndependentSource(
    space=space, angle=openmc.stats.Isotropic(), energy=energy)
```

Each configuration was run for 20 batches of 5000 particles (100,000 source histories total). TBR is scored with OpenMC’s built-in ‘H3-production’ tally score, summed over the entire geometry with no cell filter (tritium production is physically zero outside the Li-bearing sectors, so a global tally is equivalent to a sector-restricted one here):

```
tbr_tally = openmc.Tally(name='TBR')
tbr_tally.scores = ['H3-production']
```

## 7 Closed-Form Cost Model

The material cost of a given  ${}^6\text{Li}$  enrichment  $\varepsilon$  follows the same blended-cost model as the parent proposal’s full-scale cost analysis (itself derived from a published MIT S.M. thesis material-cost model [2]):

$$c_{\text{blend}}(\varepsilon) = c_{\text{base}} \cdot F_{\text{base}} + f_{\text{Li}} \cdot \varepsilon \cdot c(\varepsilon),$$

with  $c_{\text{base}} = \$12.95/\text{kg}$  (natural PbLi base price,  $F_{\text{base}} = 1$ ),  $f_{\text{Li}} = 0.681\%$  (Li mass fraction in PbLi), and  $c(\varepsilon)$  the published  ${}^6\text{Li}$ -enrichment cost curve, read at the four recipe values as  $c(0.15)=\$1,150$ ,  $c(0.30)=\$2,300$ ,  $c(0.60)=\$3,150$ ,  $c(0.90)=\$3,550$  per kg of enriched  ${}^6\text{Li}$ .

Each sector’s volume follows analytically from the  $90^\circ$  torus-segment geometry of Section 4.2:  $V_{\text{sector}} = \frac{1}{4}(2\pi^2 Rr^2) = \frac{\pi^2}{2} Rr^2 = 592,176 \text{ cm}^3$  ( $0.592 \text{ m}^3$ ), giving a sector mass of 5,921.8 kg at the PbLi density above. Table 2 reports the resulting per-sector cost by recipe; the total cost of a configuration is the sum of its four sectors’ costs.

Table 2: Per-sector cost by recipe

| $\varepsilon$ | $c_{\text{blend}}(\varepsilon)$ (\$/kg) | Sector cost |
|---------------|-----------------------------------------|-------------|
| 0.15          | 14.12                                   | \$83,643    |
| 0.30          | 17.65                                   | \$104,513   |
| 0.60          | 25.82                                   | \$152,905   |
| 0.90          | 34.71                                   | \$205,532   |

## 8 Preliminary Pipeline Checks

Before the full enumeration, three sanity-check runs confirmed the pipeline end to end (Table 3). The solid-sphere and tungsten-sphere checks used the same 14.1 MeV point/isotropic source described in Section 6 of an earlier development log; the box-sector and final torus-sector checks used the actual  $I = 4$  ring source of this report. The increasing leakage fraction and TBR from the box-sector to the torus-sector geometry (Table 3, last two rows) is the expected direction of change: the smooth, continuous torus-sector geometry presents a continuous target to the ring source, whereas the box-sector geometry leaves large gaps between sectors through which source neutrons escape without interacting.

Table 3: Preliminary pipeline verification runs

| Check            | Geometry           | Leakage fraction    | Notes                         |
|------------------|--------------------|---------------------|-------------------------------|
| Pure CSG         | 10 cm water sphere | $0.9065 \pm 0.0022$ | Confirms nuclear-data path    |
| DAGMC round trip | 10 cm W sphere     | $1.5555 \pm 0.0052$ | $(n, 2n)$ multiplication in W |
| Pipeline (crude) | 4 disjoint boxes   | $1.1154 \pm 0.0010$ | TBR = $0.0305 \pm 0.0005$     |
| Final geometry   | 4 torus sectors    | $1.6345 \pm 0.0017$ | TBR = $0.1193 \pm 0.0008$     |

## 9 Full Enumeration Results

All  $4^4 = 256$  configurations were evaluated sequentially on a single workstation (Intel, 8 logical threads), each run taking approximately 3–4 seconds (dominated by cross-section loading, not transport, at this particle count), for a total wall-clock time of approximately 12 minutes. Every configuration’s full input (recipe assignment), TBR (mean and standard deviation), and closed-form cost are archived as an individual JSON record (`results/call_0000.json`–`call_0255.json`), following the same per-call raw-trace reproducibility convention used in the parent nuclear-fuel-assembly case study [1], together with a `summary.json` containing all 256 records and the computed Pareto frontier.

### 9.1 Range of outcomes

Across all 256 configurations, TBR ranged from 0.0568 (all four sectors at  $\varepsilon = 0.15$ ) to 0.1882 (all four sectors at  $\varepsilon = 0.90$ ), and cost ranged from \$334,573 to \$822,129 over the same two extreme configurations respectively.

### 9.2 True Pareto frontier

Filtering all 256 configurations for non-domination (maximizing TBR, minimizing cost) yields exactly 25 non-dominated points, reported in full in Table 4. Two structural observations support the internal consistency of this result. First, sorting the frontier by cost produces a list that is *simultaneously* sorted by TBR with no exceptions — every point on the frontier that costs more also has strictly higher TBR than every cheaper frontier point, which is a necessary (though not sufficient) property of a correctly computed non-domination filter given that cost is a sum of four independent, monotonically-ordered per-sector terms. Second, the two extreme points of the frontier are exactly the two constant-recipe (all sectors identical) configurations at the lowest and highest enrichment, as physically expected.

Table 4: Full true Pareto frontier (25 of 256 configurations)

| Cost (\$) | TBR                 | Recipe assignment (sector0–sector3) |
|-----------|---------------------|-------------------------------------|
| 334,573   | $0.0568 \pm 0.0002$ | [0.15, 0.15, 0.15, 0.15]            |
| 355,442   | $0.0648 \pm 0.0003$ | [0.15, 0.15, 0.30, 0.15]            |
| 376,312   | $0.0724 \pm 0.0003$ | [0.15, 0.15, 0.30, 0.30]            |
| 376,312   | $0.0727 \pm 0.0003$ | [0.30, 0.15, 0.30, 0.15]            |
| 397,181   | $0.0805 \pm 0.0004$ | [0.30, 0.30, 0.30, 0.15]            |
| 418,050   | $0.0881 \pm 0.0004$ | [0.30, 0.30, 0.30, 0.30]            |
| 445,574   | $0.0942 \pm 0.0005$ | [0.30, 0.30, 0.60, 0.15]            |
| 466,443   | $0.1018 \pm 0.0006$ | [0.30, 0.30, 0.60, 0.30]            |
| 466,443   | $0.1019 \pm 0.0006$ | [0.30, 0.30, 0.30, 0.60]            |
| 493,966   | $0.1077 \pm 0.0006$ | [0.60, 0.30, 0.60, 0.15]            |
| 493,966   | $0.1078 \pm 0.0007$ | [0.30, 0.15, 0.60, 0.60]            |
| 514,836   | $0.1153 \pm 0.0006$ | [0.60, 0.30, 0.60, 0.30]            |
| 514,836   | $0.1156 \pm 0.0007$ | [0.30, 0.30, 0.60, 0.60]            |
| 542,359   | $0.1213 \pm 0.0007$ | [0.60, 0.15, 0.60, 0.60]            |
| 563,228   | $0.1290 \pm 0.0008$ | [0.60, 0.30, 0.60, 0.60]            |
| 594,986   | $0.1329 \pm 0.0007$ | [0.60, 0.15, 0.90, 0.60]            |
| 611,621   | $0.1425 \pm 0.0008$ | [0.60, 0.60, 0.60, 0.60]            |
| 647,613   | $0.1445 \pm 0.0008$ | [0.90, 0.15, 0.90, 0.60]            |
| 664,248   | $0.1540 \pm 0.0009$ | [0.90, 0.60, 0.60, 0.60]            |
| 664,248   | $0.1540 \pm 0.0008$ | [0.60, 0.60, 0.90, 0.60]            |
| 700,240   | $0.1555 \pm 0.0008$ | [0.90, 0.90, 0.90, 0.15]            |
| 700,240   | $0.1559 \pm 0.0010$ | [0.90, 0.15, 0.90, 0.90]            |
| 716,875   | $0.1656 \pm 0.0009$ | [0.90, 0.60, 0.90, 0.60]            |
| 769,502   | $0.1770 \pm 0.0010$ | [0.90, 0.60, 0.90, 0.90]            |
| 822,129   | $0.1882 \pm 0.0011$ | [0.90, 0.90, 0.90, 0.90]            |

## 10 Discussion

### 10.1 Physical sanity of the results

The frontier’s overall shape — TBR increasing with cost at a decreasing marginal rate as more sectors reach the highest enrichment level — is consistent with the physical expectation that  ${}^6\text{Li}$  enrichment yields diminishing returns on tritium breeding once the dominant absorption/breeding cross sections are no longer supply-limited by natural  ${}^6\text{Li}$  abundance (7.5%). The two constant-recipe extremes anchoring the frontier, and the strict cost/TBR co-monotonicity of all 25 frontier points, are both consistent with a correctly implemented non-domination filter and a correctly parameterized cost model.

### 10.2 Limitations and open questions for expert review

We highlight the following simplifications, on which we specifically solicit feedback before scaling to the  $I = 80$  case:

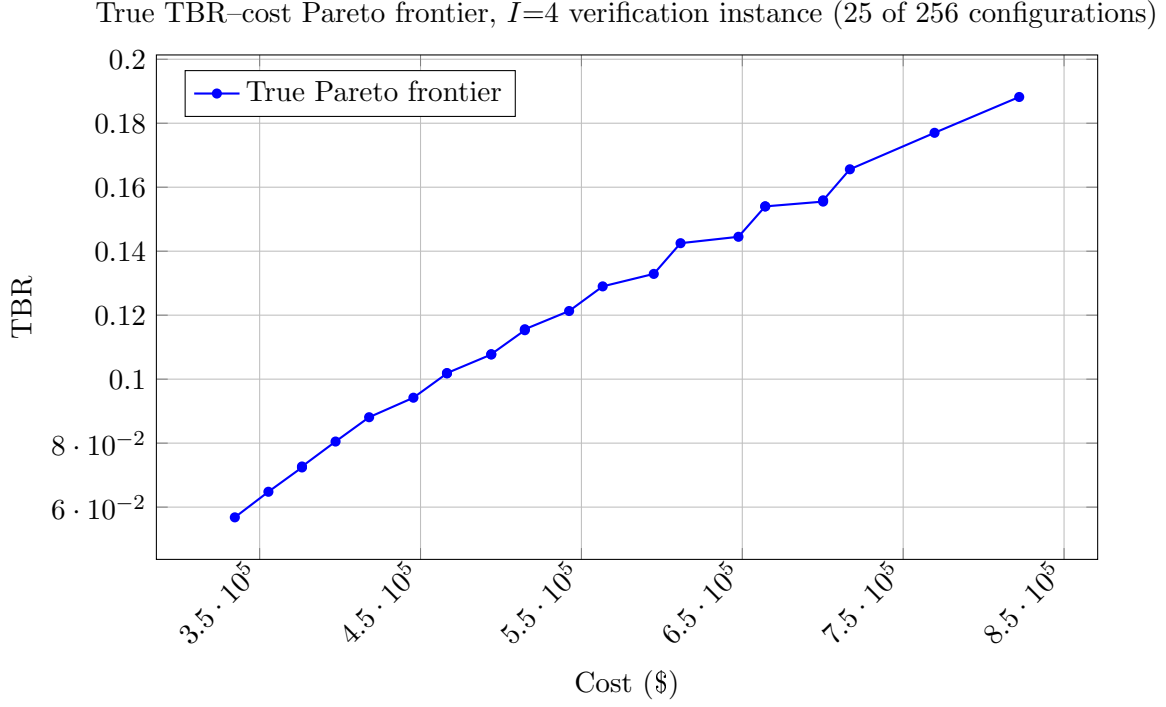


Figure 2: The 25-point true Pareto frontier out of 256 exhaustively-evaluated configurations. The monotone, near-convex shape is consistent with the cost model being a simple sum of four independent monotone per-sector terms and TBR responding smoothly to increasing overall  ${}^6\text{Li}$  content.

1. **Solid, not hollow, geometry.** The current sectors are solid PbLi torus segments with no distinct first wall, structural fraction, or neutron multiplier layer. This was a deliberate simplification to isolate and resolve the CAD/DAGMC pipeline issues documented in Section 4.2, but it is not representative of a real WCLL-style breeding blanket, and the TBR values reported here should not be interpreted as realistic predictions for any specific reference design.
2. **Unvalidated scale parameters.** The major radius ( $R = 300$  cm) and tube radius ( $r = 20$  cm) were chosen for fast iteration during pipeline development, not calibrated against any published reference geometry.
3. **Statistical precision.** 100,000 histories per configuration (20 batches  $\times$  5000 particles) gives TBR relative standard deviations of roughly 0.4–0.6% across the sweep; this is likely adequate for verifying frontier monotonicity but may be insufficient for distinguishing closely-spaced frontier points (e.g. the two points both reporting TBR near 0.1540 at identical cost) with high confidence.
4. **No neutron multiplier or structural material.** Only PbLi is present; no beryllium or  $\text{Be}_{12}\text{Ti}$  multiplier, and no Eurofer/SS316 structural fraction, both of which are part of the full-scale design described in the parent proposal.

### 10.3 Relationship to the parent proposal

This report fulfills the first half of the parent proposal’s Section 3.3.2 verification strategy: a true, exhaustively-computed Pareto frontier for a small instance. The proposal’s stated next step is to run the same surrogate-screened Adaptive Randomized Rounding (ARR) search framework used in the completed nuclear-fuel-assembly case study against this small instance, and to check its own approximate Pareto frontier directly against the ground truth reported here — rather than only against a literature-reported target, which was the strongest verification available for the nuclear fuel case. That surrogate-search comparison, and the subsequent scale-up to the full  $I = 80$  configuration space ( $4^{80} \approx 10^{48}$  configurations, for which brute-force enumeration is impossible), are left to a follow-up report.

## 11 Reproducibility

All code, the DAGMC geometry file, and the complete raw per-call results (256 JSON records plus the aggregated summary and Pareto frontier) are released in the accompanying GitHub repository. The repository structure is:

```
i4-fusion-blanket-verification/  
  README.md  
  report.tex, report.pdf <- this document  
  code/  
    i4_case_v3.py <- single-configuration reference script  
    run_full_enumeration.py <- full 256-configuration sweep  
    i4_blanket_v3.h5m <- DAGMC geometry file (shared across all 256 runs)  
  results/  
    call_0000.json ... call_0255.json <- one file per configuration  
    summary.json <- all 256 records + Pareto frontier
```

A reader can independently verify every reported TBR value directly against the corresponding `results/call_NNNN.json` file without re-running anything, or reproduce the full sweep with:

```
mamba activate fusion-env  
python run_full_enumeration.py
```

given the software stack of Table 1 and the ENDF/B-VIII.0 nuclear data library installed with `OPENMC_CROSS_SECTIONS` set accordingly.

## 12 Conclusion

We have built, debugged, and exhaustively verified a small ( $I=4$ ,  $K=4$ , 256-configuration) toroidal breeding-blanket neutronics model using an entirely open-source stack (OpenMC/DAGMC, Paramak, CadQuery, `cad_to_dagmc`), documenting both the working final approach and the CAD-construction failure modes encountered along the way so that others attempting similar DAGMC-based sector models can avoid the same pitfalls. The resulting 25-point true Pareto frontier provides a ground-truth benchmark for the surrogate-screened search comparison planned as the next step of this research line, and we welcome expert feedback on the geometry and modeling simplifications identified in Section 9.2 before proceeding to the full-scale  $I = 80$  instance.

## References

- [1] K. Chang, H. J. Park, S. Shim. *Surrogate-Screened Combinatorial Search Against an Expensive, Slow-Converging PDE Oracle: A Case Study in Nuclear Fuel Assembly Design*. Manuscript submitted to Mathematical Programming Computation, 2026. <https://doi.org/10.13140/RG.2.2.35458.64966>
- [2] J. Webber. *[Fusion breeding-blanket material cost modeling]*. MIT S.M. thesis, 2025. <https://hdl.handle.net/1721.1/165586>